PG5602 Exam Autumn 2026

Build Your World

Introduction

Places tell stories. A statue, football stadium, hiking spot, street art installation, viewpoint or favourite coffee spot can all become part of a personal geographic collection.

Your task is to build a native iOS application wherein users can ***create and manage their own geographic data***.

You will not introduce any third-party frameworks or SDKs to your solution and are expected to build what functionality is needed. Only Apple frameworks are expected to be implemented in the solution.

Your app must also connect to an **external API**, that either helps the user find places or adds useful information about the places they create.

The main content of the application must be created by the user. Each saved item must represent something that has a geographic location. Examples include but are not limited to:

- Statues, football stadiums, hiking spots, street art, restaurants, historical places, viewpoints, personal memories or maybe something completely different.

Functional Requirements

For user created items, the user must be able to:

- Create new items
- View saved items
- Open a detailed view of said item
- Edit existing items
- Delete items

You yourself decide what information a geographic item that a user creates should store, such as title, description, category, date, rating or notes etc. Items however must contain geographic data.

The data the user creates must persist its state between app launches.

External API:

The application you build must employ at least one external API that meaningfully interacts with the geographic data that is generated by the user. By meaningful interaction, we mean the following:

- Find a place (example: search for an address or place name)
- Enrich a place (example: show weather for the location)

Suitable services to complete this task may include **API's from Kartverket** such as address or place-name search, or weather data from **MET Norway** or other open API's. These API's are merely suggestions.

You are encouraged to explore **data.norge.no** for Norwegian open data and API's, but again any suitable API of your desire may be employed in their stead.

The only qualifier as mentioned above is that the API must meaningfully interact with the data created by the user.

Location:

Your application must employ the **Core Location** framework in a meaningful way. It can for example be tied to the external API you employ, the places you create or any other creative way you can find to employ it.

An example of such a meaningful integration might be:

- Save the user's current location
- Show the distance to saved objects
- Find objects close to the user
- Use the current location when creating items
- Retrieve external information for the user's location

One More Thing:

Your app must implement **at least one** additional Apple Framework that meaningfully improves the app. Some examples might be but are not limited to:

- MapKit
- PhotosUI
- AVFoundation
- HealthKit
- Foundation Models

The feature should provide real value and should not only be included to satisfy a technical requirement. Put on your product thinking caps and be creative!

The theme is up to you. You decide what is worth mapping.

Deliverables

The submission must encompass the following deliverables delivered in a zip file.

- The complete Xcode project. The project is expected to compile and run without modification.
- A video walkthrough of the application you built. Demonstrate and describe the app and its most important functionality. (max 5 minutes).
- A video of a code walkthrough. Choose 2-3 interesting parts of your implementation and explain what the code does and why you chose to perform it in this particular manner. (max 5 minutes).

Summary of Requirements

- Be built using Swift and SwiftUI
- Let the user create their own geographic objects
- Store the objects locally using SwiftData with CRUD actions
- Use Core Location + one additional Apple Framework
- Retrieve data from at least one external internet-based API
- Follow standard iOS design conventions (Human Interface Guidelines)
- Compile and run in the simulator

Assessment Criteria

Your deliverables are evaluated as a complete package across the Xcode project and the 2 videos.

Each of the following categories are assessed as such:

- Swift & Code Quality (20%)
    - Readable and idiomatic Swift, naming, structure, and appropriate use of language features covered in the course.
- SwiftUI & User Experience (20%)
    - Navigation, layout, interaction, platform conventions, clarity and overall usability.
- SwiftData & User Created Data (20%)
    - Data modelling, persistence and correct implementation of CRUD.
- Networking & External API's (15%)
    - Network communication, decoding, error handling and meaningful use of external data.
- Core Location & iOS capabilities (15%)
    - Meaningful use of location and the additional native iOS capabilities you selected.
- Product thinking & Concept (10%)
    - How well the app's features work together and the student's ability to explain their choices.

Final Note

This exam is not about building the biggest app. It is however about showing that you can combine Swift, SwiftUI, SwiftData, networking, location and the iOS platform into a focused and useful product.

Build something useful and make it yours.

Good luck!